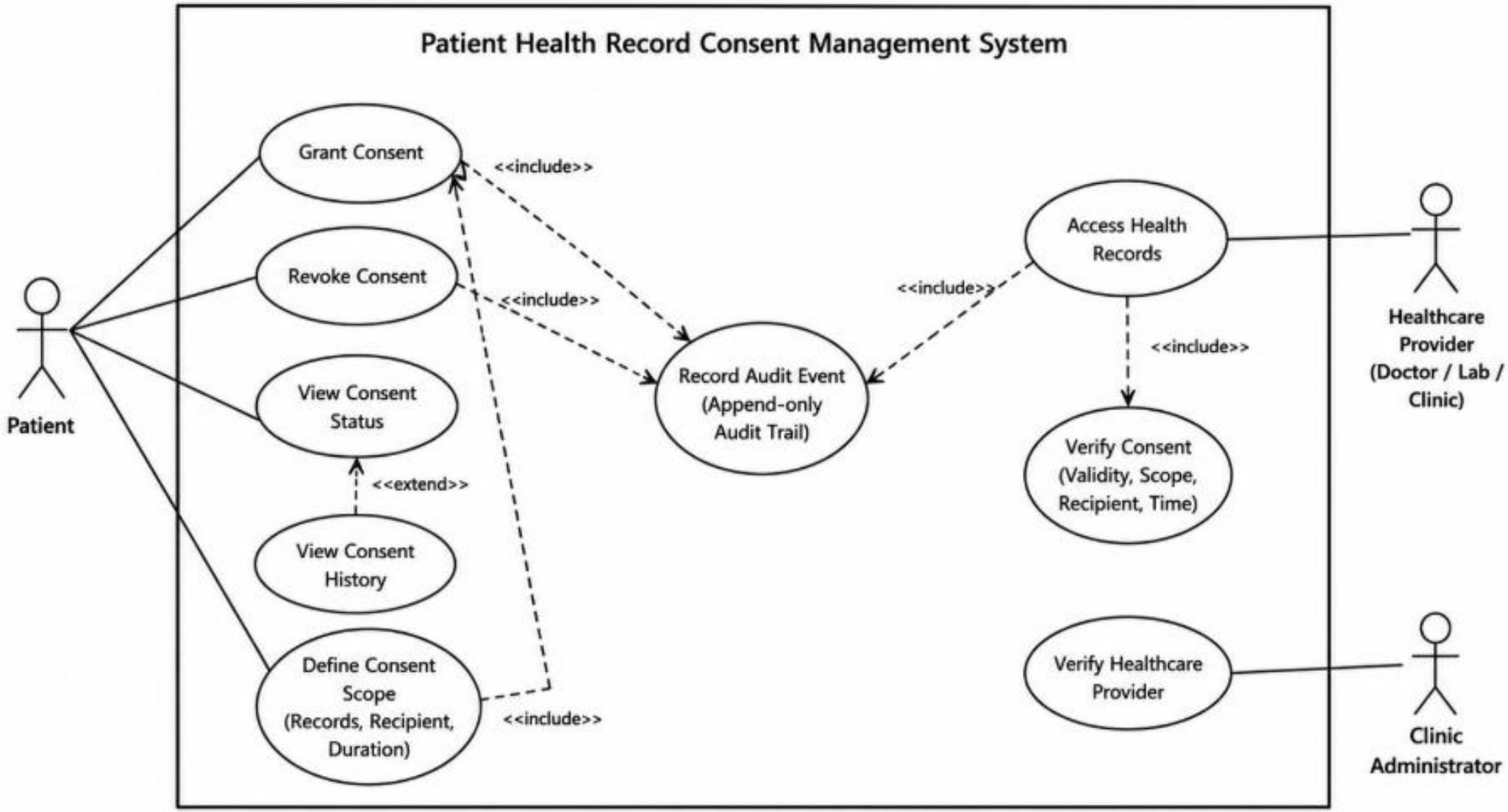
Requirement Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow patients to grant a time-bounded consent for verified clinic doctors to access specified diagnostic records.	High	Pass: Access to the selected records is permitted only until the stored consent expiry timestamp. Fail: Access remains available after the expiry timestamp.	Core requirement for patient-controlled, time-limited health-data sharing.	Specify whether the consent duration is selected from predefined options or entered as an expiry timestamp.
FR-002	Functional	The system shall allow patients to revoke an active consent before its scheduled expiry time and immediately terminate the associated access.	High	Pass: After revocation, subsequent access attempts using that consent are denied. Fail: The provider can still access the records after revocation.	Patients must be able to withdraw previously granted access without waiting for automatic expiry.	Clarify whether revocation affects only future access or also previously downloaded records.
FR-003	Functional	The system shall allow patients to define the scope of a consent by selecting the specific health records and intended verified recipient.	High	Pass: The recipient can access only the records explicitly included in the consent. Fail: Any unselected record is accessible using the same consent.	Implements the granular consent requirement rather than granting unrestricted medical-record access.	Define the supported record categories/types, such as diagnostic reports, prescriptions, or consultation records, if applicable.
FR-004	Functional	The system shall authorize a healthcare provider's record-access request by validating the provider's identity and the corresponding consent's recipient, scope, and validity period.	High	Pass: Access is granted only when provider identity, record scope, and consent validity all match. Fail: Access is granted when any validation condition fails.	Prevents a valid consent from being misused by another provider or for records outside its scope.	Clarify how verified-provider identity is represented and matched to a consent.
FR-005	Functional	The system shall allow patients to review their active and recently revoked or expired consents, including recipient, permitted records, status, and expiry timestamp.	Medium	Pass: The consent list displays the correct recipient, record scope, status, and expiry for each consent. Fail: Any displayed consent contains incorrect status, recipient, scope, or expiry information.	Gives patients visibility into who currently has access and how their previous permissions have changed.	Define the retention period for revoked/expired consent history if historical records are required.
NFR-001	Non-Functional (Security)	The system shall permanently record every consent grant, revocation, expiry, and health-record access event in an append-only audit trail.	High	Pass: Each tested event generates an immutable audit entry containing the event type and timestamp. Fail: An event is missing or an existing audit entry can be modified.	Provides accountability and supports investigation of unauthorized or disputed access.	Specify additional audit fields such as patient, recipient/provider, record identifier, and event outcome.
NFR-002	Non-Functional (Performance)	The system shall complete consent grant, revocation, and consent-validation operations within 2 seconds for at least 95% of requests under the defined test workload.	Medium	Pass: At least 95% of tested operations complete within 2 seconds. Fail: More than 5% exceed the 2-second threshold.	Consent decisions occur during health-record access, so excessive delay could interfere with clinical workflows.	Define the test workload, such as number of concurrent requests and database size, before performance testing.

Use Case Diagram



Use-Case Flow Specification

Patient Health Record Consent Management System

Goal	Grant Consent
Primary Actor	Patient
Secondary Actor	Clinic Administrator
Preconditions	1. Patient is registered and authenticated. 2. Patient's health records are available in the system. 3. The clinic is registered in the system. 4. Patient is authorized to manage consent for the selected records.
Postconditions	1. Consent is created for the selected clinic. 2. Selected diagnostic records and consent expiry time are recorded. 3. The consent-grant event is recorded in the audit trail. 4. The authorized healthcare provider can access the permitted records until the consent expires.

Main Flow	1. Patient selects Grant Consent 2. System displays the patient's available diagnostic records. 3. Patient selects the diagnostic records to be shared. 4. System displays the verified healthcare providers available for consent. 5. Patient selects the healthcare provider to receive access. 6. Patient specifies the consent duration or expiry time. 7. System validates the selected records, healthcare provider, and expiry time. 8. System displays the consent details for confirmation. 9. Patient reviews and confirms the consent. 10. System creates the consent with the selected records, clinic, and expiry time. 11. System records the consent-grant event in the append-only audit trail. 12. System confirms that the consent has been successfully granted.
Exception Flow	7.1 Invalid consent details: If the selected records, healthcare provider, or expiry time are invalid, the system displays an error message. The patient corrects the details and continues from Step 7.